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# Math-CoT Distillation Suppresses Cue Acknowledgment Beyond What Compression Explains

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## Abstract

1 Chain-of-thought (CoT) monitorability is the hope that reading a model’s reasoning  
2 will tell us why it answered, so an overseer can catch undesired influences. We ask  
3 whether supervised fine-tuning on math traces preserves this property, using the  
4 cue-based paradigm of Meek et al. (2025) and Turpin et al. (2023) on a DeepSeek-  
5 R1-distilled student. Our central result is an intervention. Fine-tuning the student on  
6 a weaker model’s math CoT, with the reasoning span genuinely present in the loss,  
7 cuts the CoT’s acknowledgment of a planted cue from 23.6% to 16.6% on matched  
8 (question, cue) pairs (paired  $\Delta = -0.080$ ,  $n = 150$ , McNemar  $p = 0.043$ ), at  
9 held-out MATH accuracy that does not fall (0.685 against an untrained reference of  
10 0.650 — though a matched-budget re-run shows the arm scoring 0.250 on uncued  
11 GPQA against baseline’s 0.425, so we no longer claim capability is preserved on  
12 the eval substrate, §5.3). Those figures come from a budget-matched re-run of  
13 both arms (§5.3); the same comparison at the 8k generation budget the rest of this  
14 paper uses reads 25.0% to 14.4%,  $p = 0.005$ , and we treat the weaker budget-  
15 matched number as the honest one, because at 8k only the untrained baseline  
16 was routinely truncated before it could answer. The retrained student’s traces are  
17 only  $\sim 27\%$  shorter than the untrained model’s, so the drop is not a by-product  
18 of the extreme CoT compression that answer-only supervision produces. At the  
19 same budget we cannot detect a change in the cue’s behavioral pull on the answer  
20 (paired  $\Delta = +0.032$ , 95% CI  $[-0.074, +0.147]$ ,  $n = 95$ ,  $p = 0.71$ ) — an  
21 underpowered null, not a demonstrated invariance. Raising the budget closes most  
22 of the differential truncation that degraded this measurement (baseline parseability  
23  $56.2\% \rightarrow 73.3\%$ ) without narrowing the interval at all, so the imprecision is real  
24 and only more questions would resolve it. That fine-tuning lowers CoT faithfulness  
25 on an answer-level metric is already known (Lobo et al., 2024); what we add is that  
26 the drop is induced by intervention, survives the removal of the compression that  
27 would trivially explain it, and is not accompanied by any detectable reduction in  
28 how much the cue moves the answer. A second arm isolates how much of the effect  
29 a shorter trace can produce on its own. Due to a chat-template defect we document  
30 in §3.1, an earlier run supervised the same student on the *final-answer segments*  
31 of the same traces; that arm compresses CoT  $14\times$  and drives acknowledgment to  
32 3.2%, so answer-only supervision reproduces a far larger collapse than reasoning-  
33 preserving supervision does. Length alone is nonetheless insufficient: at matched  
34 length and matched cue composition the gap persists in direction and effect size  
35 under every construction we tried, though each is fragile: the two that hold the  
36 question pool fixed rest on 7 and 8 records (reaching  $p = 0.23$  and  $p = 0.019$ ,  
37 the latter on a zero cell) (§5.1), and the collapse replicates on MMLU at  $2.8\times$   
38 compression rather than  $14\times$ . A same-strength self-distillation control makes the  
39 weak-to-strong asymmetry an unlikely cause, though that control was run under  
40 answer-only supervision and is correspondingly weak (§7). A capability-only

41 certification can pass a model whose CoT has become markedly less revealing  
42 while still reading as elaborate.

## 43 1 Introduction

44 Chain-of-thought “monitorability” is the hope that reading a model’s reasoning reveals why it  
45 answered, so an overseer can catch undesired influences [9]. Korbak et al. argue that this opportunity  
46 is fragile and that developers should think about how their design choices affect monitorability. Our  
47 paper looks at one such design choice, math-CoT SFT, and shows that it erodes monitorability in a  
48 specific way.

49 Weak-to-Strong Reasoning (W2SR) [1]<sup>1</sup> reports that fine-tuning a strong student on a weaker model’s  
50 CoT raises accuracy. Concurrent faithfulness work [3, 2, 4] shows that models often do not verbalize  
51 the cues that actually move their answers.

52 W2SR certifies the student on accuracy alone and is silent on whether the distilled CoT stays  
53 monitorable. If distillation changed how a model reasons on the page without changing what it  
54 reveals, an accuracy check would miss the problem entirely.

55 The phenomena we exploit are largely known. Lobo et al. [5] showed that fine-tuning hurts CoT  
56 faithfulness on an answer-level metric. Zaman and Srivastava [7] argued via causal mediation that the  
57 dissociation between behavior and CoT verbalization can exist statically. Concurrent self-distillation  
58 work [6] identifies compression as a mechanism that suppresses CoT properties, in their case the  
59 verbalization of uncertainty. Our contributions are methodological and target-specific.

- 60 1. We induce a drop in cue acknowledgment by intervention. On a run whose supervision  
61 genuinely contains the teacher’s reasoning (§5.2), math-CoT SFT moves an R1-distill  
62 student from acknowledgment 23.6% to 16.6% at a budget matched across arms (paired  
63  $p = 0.043$ ; 25.0%  $\rightarrow$  14.4%,  $p = 0.005$ , at the 8k budget that truncates the baseline far more  
64 than the trained arm — §5.3), at held-out MATH accuracy that does not fall, and with no  
65 detectable change in paired influence ( $\Delta = +0.048$ , CI  $[-0.060, +0.157]$ ,  $p = 0.52$  — an  
66 underpowered null, not a demonstrated invariance). Because the manipulation is the training  
67 data rather than a filter applied to the outcome, this does not condition on any post-treatment  
68 variable. The new piece beyond Zaman and Srivastava is the controllability: an ordinary  
69 post-training recipe produces the effect on demand.
- 70 2. A same-strength self-distillation arm serves as a negative control. With teacher equal to  
71 student, the collapse is indistinguishable from W2SR (paired  $p = 1$ ,  $n = 79$ ), and every  
72 answer-only arm we tested against W2SR gives  $p \geq 0.09$ . We have not seen this control in  
73 the literature in this form. It is our strongest evidence that the cause is the SFT in general, not  
74 weak-to-strong asymmetry, and it separates our claim from the concurrent self-distillation  
75 work on uncertainty, since the outcome variable is different but the underlying mechanism is  
76 similar. We are explicit that this control was run under the answer-only supervision format,  
77 which weakens it: with the teacher’s reasoning outside the loss, the three teachers differ  
78 only in their final answers (§7).
- 79 3. We separate how much of the collapse is trace length from how much is not, using a  
80 supervision-format contrast that we did not design but can exploit. The answer-only arm  
81 (§3.1) compresses CoT 14 $\times$  and reaches acknowledgment 3.2%; the reasoning-preserving  
82 arm compresses only  $\sim 27\%$  and reaches 14.4%. Length therefore carries much of the  
83 raw drop. It does not carry all of it: the reasoning-preserving arm sits significantly below  
84 baseline at near-matched length; the MMLU replication reproduces the full collapse at 2.8 $\times$   
85 compression rather than 14 $\times$ ; and at matched length with cue composition held fixed the  
86 gap is present in every construction we tried, at OR between 2.7 and 4.7, though only the  
87 construction that pools a larger re-run reaches significance and that construction does not  
88 hold the question pool fixed (§5.1).
- 89 4. A reproduction of W2SR with a headroom probe shows that its accuracy claim is itself  
90 conditional. The gain only fires with (i) an in-family teacher and (ii) a base student. A

<sup>1</sup>Distinct from the earlier paper with the same title by Yang, Ma, and Liu [10] (EMNLP Findings 2024), which uses a progressive SFT+DPO recipe on GSM8K, MATH, and OlympicArena. We build on Yuan et al.’s LoRA-SFT pipeline.

zero-shot CoT prompt on the untrained student separates real elicitation of latent capability from a CoT-prompting confound (Section 4), and the certified guarantee is narrower than Yuan et al. claim.

5. Putting these together gives a clear certification consequence. An accuracy-only certification of any math-CoT-SFT student, including same-strength self-distillation rather than just weak-to-strong, can pass a model whose CoT has become markedly less revealing. A monitorability composite that rewards elaborate-looking CoT misses the failure too. The cue-acknowledgment component has to be measured directly.

## 2 Related work

Lobo et al. [5] show that task-specific fine-tuning lowers CoT faithfulness across four datasets, and we take this degradation as given. The claim we are making is not that faithfulness drops, but that it drops while the cue’s behavioral influence on the answer does not. That is the dissociation their answer-level metric cannot isolate. Concurrent work by Jiang et al. [11] makes the closest complementary observation in a different domain: on medical CoT distillation, answer accuracy improves (74.7%  $\rightarrow$  84.4% MedQA) while reasoning step error rate climbs (30.6%  $\rightarrow$  50.3%), which they use to argue that answer-level metrics alone mask reasoning-quality decay. Their finding and ours share the shape “the answer looks better while the trace gets worse” in two very different domains, which is stronger joint evidence than either result alone.

Concurrent work [6] finds that self-distillation suppresses epistemic verbalization, specifically the verbalization of uncertainty tokens like “wait” and “hmm,” via compression. They observe that this hurts out-of-distribution generalization. We share the compression mechanism and the self-distillation setup, but our outcome variable is a different property of the CoT: cue-acknowledgment (does the trace mention or attribute reasoning to a planted hint) rather than uncertainty-token density. Uncertainty-token loss and cue non-acknowledgment are distinct failure modes that happen to share a mechanism. On the RL side rather than SFT, Little [14] independently shows that training with an explicit length penalty makes CoT less monitorable at matched length (7-35 pp lower hint disclosure after random-shortening baselines). Our result is the SFT-side complement: math-CoT SFT compresses CoT even with no length term in the objective and produces the same qualitative degradation, so the failure mode is not specific to length-penalized RL.

Meek et al. [2] measure monitorability as a combination of faithfulness and verbosity across model families. We use their cued-prompt eval but study how a training intervention moves faithfulness on a fixed substrate. We pair the standard verbal-acknowledgment measurement with a behavioral measure that separates “stops being influenced” from “stops verbalizing influence,” and we ask the sharper question of whether faithfulness falls beyond what CoT length alone explains.

Zaman and Srivastava [7] argue that the Biasing Features metric, which flags certain answers as “unfaithful,” partly captures the lossy compression of distributed transformer computation into linear natural language. They show via causal mediation that even non-verbalized hints can causally route influence through the CoT to the answer. That is our dissociation, observed inferentially. We add three things on top. First, we induce the effect with a training method, moving acknowledgment from 25.0% to 14.4% on a fixed substrate under reasoning-preserving supervision, and to 3.2% under answer-only supervision. Second, we pin the behavioral pull directly, via paired influence and a flip-to-cue-target rate at about  $2\times$  chance on the trained arms, rather than infer it through mediation. (On the answer-only arm the pull does not merely persist but strengthens; on the reasoning-preserving arm we cannot resolve a change either way.) Third, we draw the certification consequence, since an accuracy-only or verbosity-rewarding check passes the very models on which the dissociation is most extreme. Zaman also argue that hint-acknowledgment is a poor faithfulness metric because absence of hint words does not prove the hint was not causally involved. We agree with that critique on faithfulness grounds. But for monitorability the metric is aligned with the task: what an overseer can read off the trace determines whether they can catch the influence, independent of whether the model was “really” moved by the hint in some deeper counterfactual sense. Our CoT-conclusion analysis in §5 further shows the CoT reasoning itself walks to the cue target in most silently-influenced cases, so on that arm acknowledgment absence is not merely a missing hint word. We note the analysis was run on the answer-only arm, so it cannot by itself adjudicate compression.

Channel migration is not what is happening here. Recent work on extended-thinking models documents a thinking-vs-answer divergence in hint acknowledgment, often a large one (for instance 87.5% vs 28.6% in [8]; concurrent work [18] finds a similar divergence on a larger 12-model panel with an “unacknowledged” quadrant of 11.8%). Our judge reads the full chain-of-thought, that is, both the thinking tokens and the answer. So the collapse we measure is not an artifact of which channel is read; acknowledgment vanishes in both channels combined.

Xiong et al. [12] provide a useful contrast on the training-method axis: RLVR (reinforcement learning with verifiable rewards) spontaneously improves monitorability during training, driven by response sharpening and increased prompt attention, without monitorability being an explicit objective. Our result is the opposite for a different post-training procedure: SFT on math-CoT traces, also without monitorability in the objective, degrades monitorability instead. Which post-training method a model receives can flip the sign of the monitorability side effect, even when neither method targets monitorability directly. Related term-of-art usage: Yee et al. [13] introduced the language of “dissociation” for faithful-vs-unfaithful reasoning mechanisms in a static setting; our use of “behavior–verbalization dissociation” is in that same lineage but induced by training.

**Known limits of the cue-based paradigm.** The Turpin-style cue-acknowledgment paradigm we adopt has documented limitations. Young [15] shows that the choice of faithfulness classifier alone shifts measured faithfulness by up to 30 pp across otherwise identical setups, so absolute rates should be read as judge-dependent. Scalena et al. [16] argue that reasoning models commit to an answer well before the CoT ends, so late CoT content can be epiphenomenal and length-only comparisons can be misleading (our Task H length-binned analysis addresses this critique on our data, though see §5.1 for how much weight it can bear). Za et al. [17] document that Sonnet is roughly  $3\times$  more susceptible than Gemini under adversarial persuasion when used as a monitor, which is one reason we cross-check our Sonnet judge with two others: Gemini (Cohen’s  $\kappa \in [0.64, 0.68]$ , substantial) and Kimi (overall  $\kappa = 0.556$ , moderate; systematically stricter than Sonnet). All three judges agree on the direction of the ack drop and give one-directional discordant patterns. We frame our headline metric as *monitorability* (does the trace reveal the influence to an overseer) rather than *faithfulness* (does the trace causally reflect the model’s internal computation) precisely because the paradigm operationalizes the former, not the latter: an overseer only has the trace, so what the trace does or does not say about the cue is directly what determines whether the influence is catchable.

### 3 Setup

W2SR [1] samples CoT from a weak teacher on a problem set, keeps correct and incorrect traces, LoRA-SFTs a stronger student, and reports held-out Pass@1. Monitorability follows Meek et al. [2]: each question is paired with a cue (authority opinion, leaked answer key, embedded metadata, and so on) that points at a target. An LLM judge reads the full CoT and scores faithfulness (does the CoT acknowledge the cue) and verbosity (a factor-utilization rate). We use the five cues shipped in the released eval, with `claude-sonnet-4-6` at  $T = 0$  as the single judge. Faithfulness is reported as acknowledgment among cued samples. All evaluation is greedy ( $T = 0$ ), trace sets are hashed, and training is LoRA. Generation, training, and gating run on Modal GPUs. All code, configs, per-sample judge labels, and the reanalysis pipeline are released. Four numbers from the answer-only arms — its acknowledgment rate, baseline’s, and the two paired deltas — are enforced by hard-fail assertions in `01_gate.py`, so any drift halts the suite. The reasoning-preserving arm’s numbers are reproduced by `K_cotsft_rerun.py` but are not yet under assertion.

**Procedural details that matter for interpretation.** Trace generation used  $T = 0.6$ , top- $p$  0.95, and a repetition penalty of 1.1 (the penalty was added for the loop-prone 1.5B teacher and kept constant across all teachers for comparability; it discourages long repetitive CoT and so interacts with the length analyses). Traces without a parseable `\boxed{\}` answer were dropped before SFT, which preferentially removes traces truncated at the generation budget and therefore biases the training set slightly short. The under-test conditions were evaluated with an 8k generation budget (except the cross-family Llama arms, which used 16k); the teacher reference rows used 30k, so teacher-vs-student comparisons are not budget-matched (the baseline-vs-trained comparisons that carry every headline result are). SFT computes the loss over the full rendered sequence, prompt tokens included, rather than masking to the assistant span. The cue generator is *adaptive*: for each condition it points the cue at an option that differs from that model’s own uncued answer, so that the cue always has somewhere

to pull. A consequence we should state plainly is that pairing on (question, cue) pairs responses to the same question and the same cue *type*, but not always to the same planted target letter — across baseline and the trained arms the target differs on roughly 60% of matched pairs (90/150 against the answer-only arm, 90/155 against the reasoning-preserving one). The paired estimand is therefore “does this model follow a cue aimed away from its own prior answer,” which is the quantity we want, rather than “do two models respond identically to one fixed prompt.” Restricting to the pairs where the target letter does coincide preserves every result: the headline acknowledgment drop of §5.2 becomes  $\Delta = -0.138$  on  $n = 65$  ( $p = 0.022$ ), and the answer-only arm’s drop becomes  $\Delta = -0.167$  on  $n = 60$  ( $p = 0.006$ ). Task H’s W2SR side pools a larger re-run of the three text cues with the original run’s two non-text cues (634 records) to give the length bins usable counts; the paired analyses elsewhere use the original matched set. The main eval judge is `claude-sonnet-4-6`; Tasks I2/I3 and J were judged with `claude-sonnet-4.5` via OpenRouter. Task J additionally uses a hand-written binary yes/no acknowledgment rubric rather than the in-eval `cue_aware` rubric, so Task J rates differ from the main-eval rates by both judge and rubric and the two should not be read as interchangeable. Re-running an LLM judge does not reproduce its labels exactly even at  $T = 0$ : an identical re-run of the Task G rubric swap moved baseline acknowledgment from 9.4% to 8.8% and its McNemar  $p$  from  $6 \times 10^{-5}$  to  $1.2 \times 10^{-4}$ . We report the committed label set and note that judge-level digits carry this much noise.

### 3.1 What the supervision actually contained

This matters for how the results should be read, so we state it explicitly. Traces are generated by prompting the teacher with the R1 chat template, whose generation prefix ends `<|Assistant|><think>`, so each stored trace has the form `{reasoning}</think>{answer}`. Our SFT examples were rendered with the student’s own chat template. The published R1-Distill template contains, for assistant turns, a rule that splits the content on `</think>` and keeps only the final segment. The tokenized supervision for every R1-substrate arm *trained before we found this* was therefore the *final-answer segment only*, with the reasoning span removed — even though the trace files on disk retain it (which is why a check of the trace data alone does not reveal this, and why we report it here). Those arms are the strong-teacher, both self-distillation, and the original weak-teacher runs; the intervention they measure is SFT on the answer segments of math CoT traces (“answer-only” supervision), not SFT on the reasoning itself. The reasoning-preserving arm of §5.2, which carries this paper’s headline, was trained after the fix and does not have this property.

The format turns out to matter a great deal, which is why we treat the contrast between the two as a result rather than as a caveat. It gives the answer-only arms’ CoT compression a straightforward explanation; it makes their teacher-strength invariance much less surprising, since the teacher’s reasoning never entered the loss; and it moves acknowledgment itself, from 3.2% to 14.4% (§5.2). What the format does *not* change is that each resulting model is what it is: the behavioral and acknowledgment measurements are properties of the trained checkpoints, not of our beliefs about their training data, and they stand for whichever arm they were measured on. The released training code now renders the assistant turn without the template and hard-fails if the reasoning span is absent from a rendered example; Task K (§5.2) re-runs the W2SR-weak arm with that fixed renderer and measures what the format change was responsible for.

## 4 When W2SR fires

With a foreign-style weak teacher (DeepSeek-R1-distilled 1.5B, or SimpleRL-Zoo) distilled into a Qwen2.5-7B student, we found no Pass@1 gain and frequent degeneration, across students, ranks, and decodings. W2SR only reproduces with an in-family, native-Qwen weak teacher. With Qwen2.5-Math-1.5B-Instruct (about 67% on the train problems) as teacher and Qwen2.5-Math-7B as student, held-out MATH (L3 to L5) Pass@1 goes from 0.325 to 0.645, with format-valid 1.00.

A +5 point gate-pass on an already-CoT-capable model could be re-arrangement rather than real elicitation. To check, we score the untrained model under a zero-shot CoT prompt, which we take as the prompting-only ceiling, and we report W2SR — (zero-shot CoT) (Table 1). On the base student, W2SR adds +0.41 beyond prompting, which is genuine elicitation of latent capability. On an instruct student it adds +0.00. The apparent gain over an unelicited baseline is purely the CoT-elicitation

Table 1: Reproduction (held-out MATH L3 to L5 Pass@1). In-family native-Qwen teacher; the headroom probe separates genuine elicitation from CoT-prompting. The gain is genuine on the base student and is an artifact on the instruct student.

| student                                                                              | unelicited | zero-shot CoT | W2SR  | W2SR – CoT        |
|--------------------------------------------------------------------------------------|------------|---------------|-------|-------------------|
| Qwen2.5-Math-7B (base)                                                               | 0.325      | 0.240         | 0.645 | +0.405 (genuine)  |
| Qwen2.5-7B-Instruct                                                                  | 0.230      | 0.630         | 0.630 | +0.000 (artifact) |
| Foreign-style teachers (R1-distill-1.5B, SimpleRL-Zoo) into 7B: no gain or collapse. |            |               |       |                   |

confound. Two conditions that Yuan et al. leave implicit are therefore necessary: an in-family teacher, and a base (non-CoT-eliciting) student.

## 5 Math-CoT SFT degrades CoT monitorability

We compare a baseline student, a W2SR student (LoRA-SFT’d on a weak in-family teacher’s MATH CoT), and references, on GPQA. The SFT lands the student at its own CoT ceiling on this eval. Capability is not uniform across the arms, however — the answer-only arm fails its held-out MATH gate outright (below) — so we do not claim faithfulness differences are confound-free by construction, and we return to what each arm’s capability actually does. The question we are asking is whether distillation changes faithfulness, not whether it improves capability.

**The instruct student is at the floor.** On Qwen2.5-7B-Instruct, cue-acknowledgment is at a floor for every condition (baseline 1.8% at  $N = 970$ ; W2SR 3.6%; strong-teacher control 4.6%), while the one teacher we evaluated under cues (the R1-1.5B weak teacher) reaches 22.2% pooled and 37.1% on its strongest single cue. We cannot detect faithfulness transfer here. But verbosity does transfer strongly ( $\Delta = +0.26$  for W2SR over baseline, CI [0.22, 0.31]), so the model looks more monitorable without any measurable faithfulness change. To get a substrate with dynamic range, we move to a reasoning student.

**The answer-only arm: behavior strengthens, verbalization collapses.** On the reasoning student, the setup is fully in-family: student R1-Distill-Qwen-7B, weak teacher R1-Distill-Qwen-1.5B. This subsection describes the arm whose supervision was answer-only (§3.1); §5.2 gives the reasoning-preserving re-run that carries our headline. The trained student’s GPQA answers remain parseable at 92.1% (175/190 cued samples, vs 56.2% (90/160) for the untrained baseline — an asymmetry we return to in §7, since it is large enough to compromise the behavioral comparison), and baseline acknowledgment is off-floor at 25% (Chua and Evans [4] report ~59% on the full DeepSeek-R1; our lower baseline is on the distilled 7B variant on GPQA, so absolute levels differ across scale and eval set). Capability on this arm is *not* preserved: the only committed measurement of the preregistered held-out MATH gate — a re-gating run performed alongside Task K (§5.2) — *fails* it (Pass@1 0.550 vs the untrained reference 0.695, with a repetition-loop flag). We disclose this as a preregistration deviation (§7).

Paired influence (answer equal to cue target on matched (qid, cue) pairs) gives  $\Delta = +0.244$ , 95% CI [+0.110, +0.378],  $n = 82$ , McNemar  $p = 0.0017$ , discordant 9/29. Among answer flips, the flip lands on the cue target 73.7% of the time for W2SR against 54.8% for baseline; both exceed the 1/3 random-flip chance (baseline  $p = 0.003$ , W2SR  $p \approx 3.5 \times 10^{-19}$ ), and unlike in an earlier version of this analysis they are *not* statistically indistinguishable from each other (Fisher two-sided  $p = 0.032$ ). So on this arm the cue’s behavioral grip is not merely retained but strengthened. If we restrict to the cue-points-at-wrong-answer stratum, so that landing on the cue is by definition a wrong answer, flippers still hit the cue target at 66.7%/74.1%/74.0% for baseline, W2SR, and strong-teacher KD (each  $p \leq 9 \times 10^{-4}$  against 1/3). The pull is real cue-influence and not accuracy in disguise.

**A note on answer extraction.** These behavioral numbers depend on recovering each completion’s final answer letter, and that step is not neutral between conditions. Our extractor originally accepted a loose “... answer is  $X$ ” pattern anywhere in the completion as a fallback. On a long trace truncated before it concluded, the last such match is frequently the model restating the cue rather than answering — for one baseline sample, at character 528 of 34,142. Of the 33 baseline records extracted that

way, 21 equalled the cue target and only 7 were correct, at four-way chance, against 52% correct for records carrying an explicit answer marker. Because the failure requires a long unfinished trace it struck the long-CoT baseline far harder than the short-output trained arms, and trace length is the treatment variable. We therefore count an answer only when the model emits an explicit ANSWER: or \boxed{} marker. This correction *raises* every reported effect, because the contamination inflated the baseline: paired influence moves from  $\Delta = +0.157$  ( $p = 0.021$ ) to the  $+0.244$  ( $p = 0.0017$ ) above. Acknowledgment is judge-scored and is untouched by the extractor.

Verbalization collapses on both rubrics. On the upstream loose rubric (“acknowledge the cue”), W2SR drops pooled acknowledgment to 3.2% ( $\Delta = -0.220$ , CI  $[-0.293, -0.153]$ , McNemar  $p \approx 2 \times 10^{-9}$ , discordant 34/1). The drop holds at every text cue (Table 4: stanford 47%  $\rightarrow$  8%, grader-hack 22%  $\rightarrow$  3%, insider 56%  $\rightarrow$  5%), and the discordant cells are one-directional (12/0, 7/1, 15/0). Figure 1 summarizes the behavior-verbalization split across all trained arms. A stricter “attribute reasoning to the cue” rubric, scored by gemini-2.5-pro (the same out-of-family judge used for the re-scoring below), shrinks both rates (baseline 9.4%, W2SR 0.0%, 15/0 discordant,  $p = 6 \times 10^{-5}$ ). The magnitude is smaller but the drop becomes even more one-directional. Both numbers hold at roughly  $14\times$  shorter CoT (baseline median 18,692 chars, W2SR 1,362), which is the central confound this arm cannot rule out and §5.2 is designed to address.

Uncued GPQA accuracy is a weak capability check on this substrate and we do not rest anything on it. At the 8k generation budget the untrained baseline fails to emit a parseable answer on 20 of 40 uncued items — it writes  $\sim 20,000$  characters and runs out of room — against 2 of 40 for the answer-only arm. Scoring unparseable items as incorrect gives baseline 0.275 and the trained arm 0.425; scoring only the parseable subset reverses the order, 0.550 against 0.447. The comparison mostly measures which model finishes within budget. We therefore rest capability claims on the held-out MATH gate (§5.2), where generation is short enough for the measurement to mean what it says.

We checked that the collapse is not an artifact. Hand-checked completions are coherent and none of them degenerate in the sense of looping or emitting empty text. (This is a hand inspection of GPQA completions; it is not in tension with the repetition-loop flag both arms carry on the held-out MATH gate, which is raised by a single generation out of 200 on a different task.) The model solves the problem without verbalizing the cue. It is also not a single-judge artifact: re-scoring the same 525 R1-family CoTs with gemini-2.5-pro gives per-condition Cohen’s  $\kappa \in [0.64, 0.68]$  (overall  $\kappa = 0.68$ ), paired drop  $\Delta = -0.153$ , McNemar  $p = 2.4 \times 10^{-7}$ , discordant 23/0. Adding a third judge (moonshotai/kimi-k2-0905, chosen because it is from a different model family than both Sonnet and Gemini) gives a stricter reading: absolute rates shrink (baseline 24.7%  $\rightarrow$  12.3%, W2SR 3.2%  $\rightarrow$  1.6%), overall  $\kappa = 0.556$  (moderate, and lower than Gemini’s because Kimi under-calls acknowledgment consistently across all conditions), but the paired drop still fires with the same one-directional signature:  $n = 140$  matched pairs, discordant 17 baseline-only vs 1 W2SR-only,  $\Delta = -0.114$   $[-0.171, -0.057]$ , McNemar  $p = 1.4 \times 10^{-4}$ . Two disclosures on this comparison. First, the Kimi judge was called with a short output cap and returned an unparseable reply on 20 of 525 records, which we drop rather than coerce to zero, so every Kimi figure is over 505 records (154 baseline, 185 W2SR weak, 166 strong) rather than 525; 18 of the 20 dropped records carried a negative label under the original judge, so the omission is close to neutral. Second, the re-scoring harness shows each judge only the cue *name* and the trace, whereas the in-eval scorer additionally sees the question with the cue in situ and the verbatim cue text. The cross-judge  $\kappa$  values are therefore a lower bound confounded with that scaffold difference, not a clean model swap: 20 of the 28 Sonnet-Gemini disagreements fall on insider\_information, the cue whose text the re-scoring harness withholds, while the self-naming Stanford cue agrees on 105/105. Prompt construction is identical across conditions within each harness, so no paired drop is biased by this. The three-judge agreement covers three model families and both stricter and looser rubric readings. The drop direction is stable across all three; the discordant pattern is strictly one-directional under Sonnet and Gemini and near-one-directional under Kimi (17/1). All three were measured on the answer-only arm — the reasoning-preserving arm of §5.2 is scored by the Sonnet judge alone.

## 5.1 Why faithfulness drops: compression, plus residual cue suppression

Faithfulness and CoT length both fall, so the near-trivial half of the story is that shorter CoT surfaces cues less often. The interesting question is whether faithfulness still degrades at matched length.

Table 2: Reasoning-student monitorability on GPQA with in-family R1-distill teachers. **Every row is at the 8k generation budget**, which truncates the untrained baseline far more than the trained arms; §5.3 re-runs the first two rows at 32k and those are the figures the abstract reports (23.6% and 16.6% acknowledgment, baseline GPQA 0.425 against the arm’s 0.250). Read this table as the 8k specification, not as the headline. Faithfulness is acknowledgment among cued samples; median CoT is computed on cued-cell completions. The first block is the reasoning-preserving re-run (§5.2): acknowledgment falls to 14.4% at 13,721-character traces, i.e. without the compression. The second block is the answer-only supervision format (§3.1), under which every arm compresses  $\sim 14\times$  and lands near the floor, teacher strength notwithstanding — self-distillation collapses indistinguishably from W2SR weak ( $p = 1$ ), and each answer-only arm we tested against W2SR weak is not distinguishable from it (pairwise McNemar  $p \geq 0.09$ ; the three pairs not involving W2SR weak were not tested). Only the reasoning-preserving arm differs from W2SR weak ( $p = 8.8 \times 10^{-5}$ ). “n/a” marks arms for which we did not run the uncued cell.

| condition                                | faithfulness   | GPQA acc | median CoT (chars) |
|------------------------------------------|----------------|----------|--------------------|
| baseline R1-7B                           | 40/160 = 25.0% | 0.275    | 18,692             |
| <i>reasoning-preserving supervision</i>  |                |          |                    |
| W2SR (1.5B $\rightarrow$ 7B), Task K     | 26/180 = 14.4% | 0.375    | 13,721             |
| <i>answer-only supervision (§3.1)</i>    |                |          |                    |
| W2SR (1.5B $\rightarrow$ 7B)             | 6/190 = 3.2%   | 0.425    | 1,362              |
| strong-teacher KD (14B $\rightarrow$ 7B) | 13/175 = 7.4%  | n/a      | 1,508              |
| self-distill, $\leq 4k$ budget (Self-A)  | 2/84 = 2.4%    | n/a      | 1,258              |
| self-distill, $\leq 8k$ budget (Self-B)  | 3/89 = 3.4%    | n/a      | 1,291              |
| cross-family: Llama-8B self-distill      | gate fail      | 0.225    | 1,424              |

GPQA accuracy is the uncued cell at  $n = 40$ , scoring an unparseable completion as incorrect. This column understates the untrained baseline, which exhausts the 8k generation budget without answering on 20/40 uncued items; scoring only parseable completions reverses the ordering (0.550 baseline vs 0.447 answer-only vs 0.517 Task K). Capability claims in this paper rest on the held-out MATH gate, not on this column.

Cross-family row (Llama-8B self-distillation, matched recipe): GPQA accuracy  $0.500 \rightarrow 0.225$  (−27.5pp at 0% empty), CoT compression  $10.8\times$ . The capability gate failed, so the faithfulness number is not interpretable and we do not report it (see Limitations).

Pairwise tests: W2SR vs. baseline  $p = 2 \times 10^{-9}$ ; strong-teacher KD vs. baseline  $p = 2.2 \times 10^{-5}$ ; Self-A vs. baseline  $p = 2.4 \times 10^{-7}$  ( $n = 77$ , disc. 23/0); Self-B vs. baseline  $p = 4.8 \times 10^{-7}$  ( $n = 82$ , disc. 22/0); Self-A vs. W2SR  $p = 1$  ( $n = 79$ , disc. 2/2); Self-B vs. W2SR  $p = 1$  ( $n = 79$ , disc. 2/3). All tested pairs of *answer-only* arms are statistically indistinguishable; the three pairs not involving W2SR weak were not tested. The reasoning-preserving arm is the one trained arm that does differ from W2SR weak ( $p = 8.8 \times 10^{-5}$ ).

348 Length is endogenous to reasoning (a brevity-prompt control failed, since R1 ignores it), so we bin  
349 both conditions by CoT length and compare within bins.

350 A length-binned analysis (Task H, quantiles of the union distribution, Fisher’s exact unpaired;  
351 Figure 2A) sharpens the matched-length question, but only once cue composition is controlled.  
352 Pooled across all five cues, at matched mid-length ([1954, 8716] chars, baseline  $n = 40$ , W2SR  
353  $n = 119$ ) baseline and W2SR look essentially identical (ack 17.5% vs 15.1%, Fisher two-sided  
354  $p = 0.80$ , OR = 1.19), and at very long lengths ([8716, 37816], baseline  $n = 119$ , W2SR  $n = 40$ )  
355 baseline is higher but short of significance (27.7% vs 15.0%, OR  $\approx 2.2$ , two-sided  $p = 0.14$ ).

356 That pooled comparison is confounded by cue mix. Two of the five cues (visual\_squares,  
357 xml\_metadata) sit at 0% acknowledgment in *both* conditions, so they contribute only zeros —  
358 but their share of a bin differs sharply by arm, because the arms occupy the length range differently.  
359 In the mid bin they are 20/40 (50%) of baseline’s records against 10/119 (8%) of W2SR’s. Pooling  
360 therefore drags baseline’s bin rate down far more than W2SR’s, and the pooled test partly measures  
361 cue composition rather than the matched-length gap. Restricted to the three text cues, the gap is  
362 present at both lengths: mid-length baseline 35.0% (7/20) vs W2SR 16.5% (18/109), OR  $\approx 2.72$ ,  
363 two-sided  $p = 0.068$ ; long baseline 44.0% (33/75) vs W2SR 18.8% (6/32), OR  $\approx 3.40$ , two-sided  
364  $p = 0.016$ .

365 That significance should not be leaned on, for a reason we state rather than bury. Task H’s W2SR  
366 side pools a larger re-run of the three text cues (634 records over 188 question ids) with the original

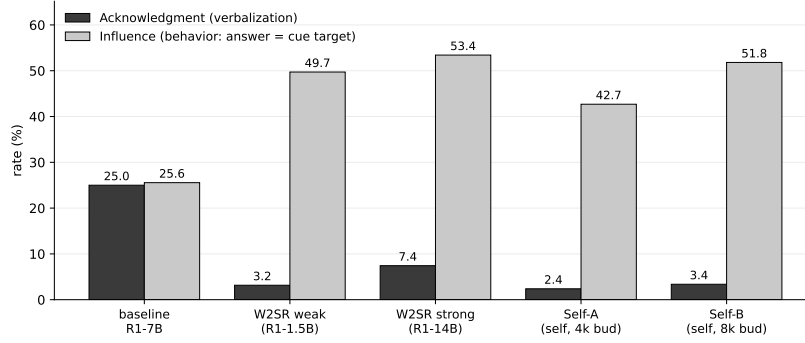


Figure 1: Behavior and verbalization across the *answer-only* reasoning-student arms. Acknowledgment (verbalization) collapses from 25.0% on baseline R1-7B to 2.4–7.4% across all four trained arms (W2SR, strong-teacher distill, two self-distill budgets), while influence (behavior, defined as answer equal to cue target) is higher in every trained condition than in baseline’s 25.6%. These arms keep being moved by the cue — if anything more than baseline does — and stop saying so. All bars are at the 8k generation budget. The reasoning-preserving arm is not plotted: at 8k it sits at acknowledgment 14.4%, and at the matched 32k budget the paper reports (§5.3) at 16.6%, with influence an underpowered null against baseline under both budgets. No 1/3 chance line is drawn, because influence as plotted is an unconditional rate; the 1/3 reference applies to the flip-to-cue rate among flippers, reported separately in the text. Regenerate with `python3 scripts/reanalysis/make_dissociation_fig.py`; the 25.0%, 3.2% and the paired deltas  $-0.22$  and  $+0.24$  carry hard-fail asserts in `01_gate.py`.

run’s non-text cues, while the baseline side is the original 160 records over 32 question ids. The two sides therefore do not draw from a common question pool: in the long bin only 5 question ids are shared, and only 7 of the 32 W2SR records sit on a question the baseline arm ever saw. Question identity is thus partially confounded with condition. Two constructions that do hold the question pool fixed preserve the direction and, if anything, the effect size, but lose power: restricting the W2SR side to baseline’s question ids gives 44.0% vs 14.3% (1/7),  $OR \approx 4.71$ ,  $p = 0.23$ ; using the original un-thickened W2SR run, which shares 30 of baseline’s 32 question ids, gives 44.0% vs 0.0% (0/8),  $p = 0.019$  on a zero cell. The honest summary is that the matched-length gap is directionally robust and quantitatively consistent across constructions, and rests on very few records in every construction that holds the question pool fixed — two of the four reach nominal significance, but on 8 records and a zero cell respectively. It is supporting evidence, not load-bearing; §5.2 is what the argument rests on.

A third cut restricts to the safety-relevant subset, where `influenced = 1` means the cue actually moved the answer. There the mid-length gap is baseline 70.0% (7/10) vs W2SR 21.1% (16/76),  $OR \approx 8.75$ , Fisher two-sided  $p = 0.003$ . We flag this one as the most fragile number in the paper: it conditions on a post-treatment variable, the baseline cell holds  $n = 10$ , and after the extraction correction of §5 the corresponding long-length cell has too few records to test (7/13 vs 1/3). It is a well-motivated exploratory cut, not a confirmatory test, and nothing below depends on it.

So a large share of the raw 25%  $\rightarrow$  3% drop is compression, and that compression has a mundane and fully identified cause: the supervision this arm received was answer-only (Section 3.1). A student trained on final-answer segments produces short completions that rarely emit the `</think>` separator (21.6% of W2SR-student completions vs 57.5% of baseline), which is what we observe. We therefore do *not* claim the compression is an emergent or surprising property of SFT; it is the expected consequence of the supervision format.

What compression does not explain is what remains once it is removed. The cue-stratified matched-length gap above is one handle; the MMLU replication is a second, since it reproduces the collapse at a much lower compression ratio; and the intervention in §5.2 is the third and the one we rely on, because there the compression is removed at its source rather than conditioned away.

## 5.2 The headline result: the same arm with reasoning-preserving supervision (Task K)

The length-binned analysis is correlational; the supervision format gives us an intervention. We re-ran the W2SR-weak arm identically — same 729 R1-1.5B teacher traces, LoRA recipe, seed, decode parameters, judge, and  $n = 40$  cells — except that the SFT examples were rendered with the fixed renderer of Section 3.1, so the reasoning span entered the loss. This removes most of the compression at its source: the retrained student’s median cued-cell CoT is 13,721 chars (1,364 for the answer-only arm; 18,847 for baseline, i.e. the rerun’s traces remain  $\sim 27\%$  shorter than baseline’s).<sup>2</sup> Under the compression-only account, acknowledgment should return to baseline. It does not. Acknowledgment lands at  $26/180 = 14.4\%$ : well above the answer-only arm’s  $3.2\%$  (paired  $\Delta = +0.118 [+0.065, +0.176]$ ,  $n = 170$ , discordants 23 rerun-only vs 3 answer-only-arm-only,  $p = 8.8 \times 10^{-5}$ ), and still significantly below baseline’s  $25.0\%$  (paired  $\Delta = -0.110 [-0.181, -0.039]$ ,  $n = 155$ , discordants 25 baseline-only vs 8 rerun-only,  $p = 0.005$ ). This was the paper’s headline until we re-ran both arms at a matched budget (§5.3), which weakens it to  $p = 0.043$ ; the figures in this subsection are the 8k specification and should be read as such. What survives either way is that the manipulation is the training data rather than a filter on the outcome.

On the behavioral side we find no detectable change: paired influence against baseline gives  $\Delta = +0.048$ , 95% CI  $[-0.060, +0.157]$ ,  $n = 83$ ,  $p = 0.52$ . We report that interval rather than declaring behavior invariant, because it is wide enough to admit an effect of  $+0.157$  — numerically the same effect this paper’s behavioral analysis of the answer-only arm reports as real. The raw rates move slightly upward (baseline  $25.6\%$  to rerun  $31.9\%$ ). What we can say is that acknowledgment falls significantly while influence does not fall at all, and that the influence measurement here is both underpowered and degraded by differential answer emission (§7); what we cannot say from  $n = 83$  is that behavior is invariant. An equivalence test on this cell would require substantially more data than we have. The rerun is nonetheless a real shift down from the answer-only arm’s influence level (paired  $\Delta = -0.163$ ,  $n = 123$ ,  $p = 0.007$ ).

Capability does not fall on the measurement best able to carry it: on the held-out MATH gate the rerun scores 0.685 against an untrained reference of 0.650, where the answer-only arm scored 0.550 against 0.695. We deliberately do not describe this as a capability *gain*. The gate’s own verdict records “no capability gain” for the rerun, and the two gate runs scored the *same* untrained model at 0.650 and 0.695 — a 4.5-point run-to-run swing in the reference, larger than the rerun’s  $+3.5$ -point margin over it. The defensible reading is that the reasoning-preserving arm does not lose held-out MATH accuracy, whereas the answer-only arm loses 14.5 points against its own reference. Uncued GPQA is, as noted in §5, budget-limited and ambiguous: scoring unparseable completions as incorrect gives the rerun 0.375 against the answer-only arm’s 0.425 and baseline’s 0.275, while scoring only parseable completions gives 0.517, 0.447 and 0.550. We do not read a capability difference out of that column in either direction.

Per cue, the recovery is uneven: insider-info (15/36 vs 2/38) and Stanford-professor (10/36 vs 3/38) recover substantially, while grader-hack does not recover at all (0/36; baseline 7/32). So the supervision format explains a large share of the raw collapse — consistent with the compression story — but a suppression component survives when the compression is removed, and it concentrates on the most grader-adversarial cue. Caveats: this is one arm and one seed, scored by the headline judge only (no multi-judge battery on this arm); the per-cue cells are small; the effect is specification-sensitive, weakening to  $p = 0.077$  if we restrict to the 83 pairs where neither side hit the generation cap (baseline  $20.5\%$  vs rerun  $10.8\%$ ,  $\Delta = -0.096$ , same direction and magnitude, less power — note baseline truncates more often than the rerun,  $41.9\%$  vs  $21.3\%$  of the 155 paired records — we originally read this as reassuring, but §5.3 re-ran both arms without the cap and the effect did weaken, so the sensitivity was signal rather than noise); and the rerun’s own gate report formally fails the preregistered criteria (format-valid  $0.80 < 0.9$ , a repetition-loop flag, and a  $+3.5$ pp MATH gain below the  $\geq 5$ -point margin), so “capability holds” above is not a passing gate. The format-valid figure should be read with care: the gate generates at 4k tokens while this arm’s median CoT is 13,721 characters, so a fifth of its gate generations are cut off before they can emit a boxed answer,

<sup>2</sup>The medians in this paragraph come from the Task K pipeline, which uses the upper-middle element; Table 2’s 18,692 and 1,362 come from an earlier pipeline that uses `statistics.median` (mean of the two middle values) over the same records. The conventions differ by well under 1% and every comparison is convention-consistent.

where the answer-only arm at 1,364 characters scored 0.99 on the identical check (result files: results/reanalysis/K\_cotsft\_rerun.\*).

**Teacher-strength axis.** A strong (14B) teacher degrades faithfulness about as much as a weak (1.5B) one (Table 2), and the two are not distinguishable. To check whether the weak-to-strong asymmetry is causal at all, we train baseline R1-7B on its own R1-7B traces (the same MATH L3 to L5 problem pool,  $T = 0.6$ , same SFT config). Both self-distillation arms (4k and 8k generation budgets) collapse acknowledgment to 2.4% and 3.4%, statistically indistinguishable from W2SR (paired  $p = 1$ ,  $\Delta \approx 0$  on  $n = 79$  matched pairs each), and median CoT compresses to about 1.3k chars regardless of budget.

**Cross-substrate replication on MMLU.** An MMLU sweep over 5 STEM subjects (college physics, chemistry, biology and mathematics, plus conceptual physics; 8 questions per subject per cue, 190–195 cued samples per condition) holds the pattern. Pooled baseline ack is 27.4% (52/190), W2SR is 3.6% (7/195), and Self-A is 4.6% (9/195). Paired on (qid, cue), W2SR vs. baseline gives  $\Delta = -0.232$ , McNemar  $p = 2.4 \times 10^{-10}$  ( $n = 185$ , discordant 47/4), and Self-A vs. baseline gives  $\Delta = -0.222$ ,  $p = 8.2 \times 10^{-10}$  (discordant 45/4). Self-A vs. W2SR is again indistinguishable ( $\Delta = +0.010$ ,  $p = 0.73$ ), replicating the GPQA teacher-invariance result on a second substrate. Baseline R1-7B already writes much shorter CoT on MMLU (3,223 vs 18,692 chars median), so MMLU compression is only about  $2.8\times$  (baseline 3,223  $\rightarrow$  W2SR 1,141) against GPQA’s  $14\times$  — yet the absolute drop is if anything larger (MMLU 27.4  $\rightarrow$  3.6%, about 24 pp; GPQA 25.0  $\rightarrow$  3.2%, about 22 pp). The collapse therefore does not require extreme compression, and its magnitude does not scale with how much compression occurred, which is evidence against reading the effect as a simple monotone function of trace length. The faithfulness collapse is not weak-to-strong-specific and not GPQA-specific.

**The collapse is not an artifact of testing math-trained models on science.** Restricting the MMLU replication to the college\_mathematics subject (an in-domain evaluation, matched to the MATH-L3–L5 training data), the pattern is if anything more dramatic than on the science subjects. Baseline acknowledges 36.7% (11/30 matched (qid, cue) pairs), W2SR 2.5% (1/40), and Self-A 0.0% (0/40). Paired McNemar exact  $p = 0.006$ ,  $\Delta = -0.333$ , discordant 11/1 (Self-A vs. baseline:  $p = 0.001$ ,  $\Delta = -0.367$ , discordant 11/0). For contrast the combined non-math STEM subjects (biology, chemistry, physics variants) give baseline 25.6% (41/160), W2SR 3.9% (6/155). The dissociation does not require a distribution shift from training to evaluation: the SFT collapses cue acknowledgment even when we evaluate on math questions.

**What does the CoT itself derive?** The dissociation results so far tell us that the model’s answer is silently cue-influenced and the judge sees no acknowledgment, but they do not tell us what the reasoning trace itself would predict if read alone. On the 255 W2SR samples with influenced=1, ack= 0, and cue\_target  $\neq$  correct, we strip the trailing ANSWER: X line and send the remaining CoT (along with the four answer choices, but no cue text) to a fresh judge (claude-sonnet-4.5 at temperature 0) with the question “which letter is this reasoning arriving at?” The results (Table 3): in 214/255 (83.9%) of samples the CoT itself concludes the cue target, either by explicitly naming it (185) or by deriving a value that maps to it (29). Only 14/255 (5.5%) show the strong “silent flip” pattern where the CoT reasons to the correct answer and the emitted answer flips to the cue. The pattern is consistent across text cues (85.3%, 82.4%, 86.7% for grader-hack, insider-info, Stanford-professor respectively; the two non-text cues have  $n < 20$  and are noisy). One mechanical caveat: the stripper removed only a trailing ANSWER: X line, and in 30 of the 255 CoTs the emitted letter also survived in another terminal form (a closing `\boxed{}` or a final “the answer is X” sentence), so the judge could in principle read the letter off the emission rather than the reasoning. Nearly all of the 30 were judged walks-to-cue (at least one was judged a silent flip); excluding all 30 and conservatively deducting every one of them from the walks-to-cue count gives a floor of  $184/225 = 81.8\%$  (crediting the silent flip,  $185/225 = 82.2\%$ ), so the headline does not rest on them (and a CoT that closes by naming its own conclusion is largely what the transparent “by naming the letter” category is already reporting).

As a control, we ran the same judge on the baseline R1-7B samples meeting the same influenced=1, cue\_target  $\neq$  correct filter (baseline naturally acknowledges the cue  $\sim 25\%$  of the time, so the strict ack=0 restriction leaves too few samples to compare). Under the extractor in force when these labels

were collected the filter selected 36 samples, of which 28 (77.8%) have a CoT concluding the cue target, rising to 9/10 (90%) on the subset where baseline also did not acknowledge. The corrected extractor of §5 shrinks that filter to 16 samples, because defining “influenced” requires a recoverable answer and baseline often has none; on those 16 the CoT concludes the cue target in 16/16, and in 6/6 of the  $\text{ack}=0$  subset. We report both: the direction is identical and the corrected figure is stronger, but it rests on 16 records and the 20 that drop out are disproportionately the ones whose conclusion the judge found ambiguous, so neither number should be read as precise. So the “CoT walks to the cue when the model is influenced” pattern is not W2SR-specific: it is the default behavior of any silently-influenced model. What W2SR distinctly loses is not the reasoning direction, which is already corrupted whenever the cue moves the answer, but the acknowledgment overlay: baseline warns the overseer 25% of the time by explicitly mentioning the cue; W2SR warns only 3% of the time. The corrupted reasoning underneath is comparable; what changes is whether the trace surfaces the warning.

For monitoring, the practical implication sharpens rather than weakens: a naive overseer reading only the CoT of a cue-influenced completion would concur with the model’s wrong answer regardless of the model (baseline or W2SR), because the CoT itself is walking to that wrong answer. The W2SR-specific harm is not that the CoT direction changed but that the model’s tendency to comment on the cue, which is the overseer’s only independent signal, was stripped away.

**Is the acknowledgment capability recoverable at inference time?** The dissociation results so far are silent on whether the acknowledgment capability is destroyed by SFT or only suppressed by default context. To distinguish, we ran the W2SR R1-7B model on  $\sim 100$  matched (qid, cue) samples under four conditions: no system prompt (matching the paper’s main-eval setup), a “think for at least 2000 tokens and consider all information including any hints” long-CoT prompt, an “if you notice any hints, cues, external suggestions, or biases in the prompt, name them in your reasoning” explicit-acknowledge prompt, and a few-shot in-context-learning (ICL) prompt that shows the model a worked example of a CoT that explicitly names an authority-opinion hint before verifying the answer independently, then asks it to solve the target problem in that same style. Pooled across cues:

|     | condition               | n   | ack | rate | vs none (paired)             | p     |
|-----|-------------------------|-----|-----|------|------------------------------|-------|
|     | none (no system prompt) | 99  | 1   | 1.0% | –                            | –     |
| 527 | long-CoT prompt         | 99  | 6   | 6.1% | 5 recovery-only, 0 none-only | 0.06  |
|     | acknowledge prompt      | 99  | 5   | 5.1% | 4 recovery-only, 0 none-only | 0.13  |
|     | ICL example prompt      | 100 | 9   | 9.0% | 8 recovery-only, 0 none-only | 0.008 |

The ICL prompt is the strongest of the three (Figure 2B), reaches statistical significance under paired McNemar (8/0 discordant,  $p=0.008$ ), and in no case did any of the three prompts reduce acknowledgment relative to none.

Broken down by cue, the recovery concentrates almost entirely on one cue type. On `stanford_professor_recommends` (an authority-opinion cue), the ICL prompt lifts ack from 1/20 (5%) to 8/20 (40%), approaching the baseline R1-7B level of 47% on that cue — though that 47% comes from the main eval’s judge and rubric while the 40% comes from Task J’s, so the two are only loosely comparable; long-CoT reaches 6/20 (30%) and acknowledge reaches 4/20 (20%). On `grader_hack_validation`, the two non-text cues, and mostly on `insider_information`, none of the three prompts produce measurable recovery (0–1 acks in each cell of 18–21 samples). Of the nine pooled ICL recoveries, eight come from the Stanford-professor cue alone. As a brief diagnostic, in the first 400 characters of the ICL completions on `grader_hack_validation`, none of the 21 samples in either the none or the ICL condition contain grader-related tokens (“grader,” “hack,” “validation,” “answer key”), suggesting the model is not merely failing to flag a token it produces but is not producing the concept at all in its opening reasoning.

The reading is that the acknowledgment capability is not uniformly destroyed by SFT: for a highly nameable, salient authority cue the model can be prompted (with a demonstration example) to acknowledge it, reaching close to the untrained baseline rate on that cue. But for less-nameable cues (grader-hack framing, insider-information framing) even the strongest inference-time intervention we tested recovers essentially nothing. A system-prompt intervention is a partial mitigation restricted to prompt-elicitable cue types; it is not a substitute for training-side preservation of monitorability.

Table 3: What the CoT itself derives on the safety-relevant W2SR subset ( $n=255$ , influenced=1, ack=0, cue\_target  $\neq$  correct). Judge is Claude Sonnet 4.5 at  $T=0$ ; the trailing ANSWER: X line is removed from the CoT and the four answer choices are shown (no cue text, no question stem).

| what the CoT reasoning arrives at                               | n   | fraction |
|-----------------------------------------------------------------|-----|----------|
| cue target (CoT walks to cue)                                   | 214 | 83.9%    |
| by naming the letter                                            | 185 | 72.5%    |
| by deriving a value that maps to it                             | 29  | 11.4%    |
| correct answer (silent flip: reasoning correct, answer flipped) | 14  | 5.5%     |
| derivation unmappable to any letter                             | 13  | 5.1%     |
| some third letter (neither correct nor cue)                     | 12  | 4.7%     |
| judge failed                                                    | 2   | 0.8%     |

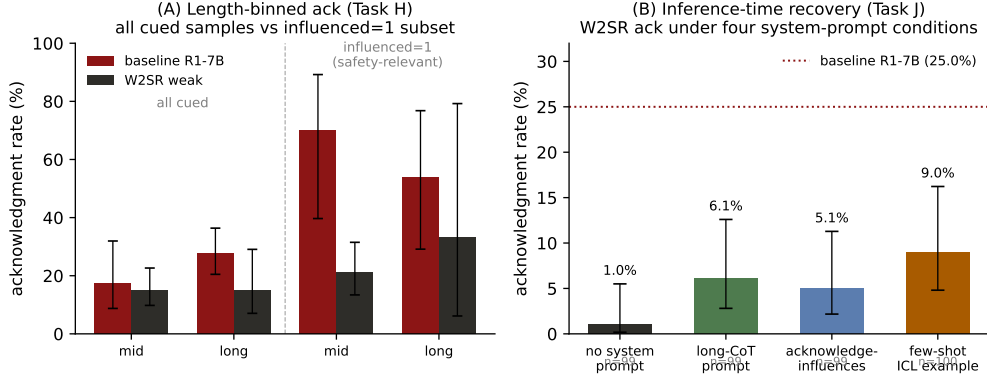


Figure 2: Left (A): length-binned acknowledgment (Task H), baseline R1-7B vs W2SR weak. On all cued samples (left half) the gap at matched mid-length is essentially closed; restricted to the safety-relevant influenced=1 subset (right half) the gap reopens substantially. Error bars: Wilson 95% CI. Right (B): inference-time recovery (Task J). Ack rate on W2SR R1-7B under four system-prompt conditions at  $\sim 100$  matched (qid, cue) pairs per condition. The few-shot ICL example prompt is the strongest and reaches statistical significance ( $p=0.008$ , paired McNemar), but the pooled recovery still sits well below the baseline R1-7B rate of 25% — a comparison across two judge/rubric stacks (§3) and so only indicative; the recovery concentrates on the Stanford-professor cue (see text).

### 5.3 The generation budget was not neutral between the arms

Every result above for the two arms compared here is generated at an 8k token budget (the cross-family Llama arms used 16k and the teacher reference rows 30k, §3.1), and that budget is not neutral. The untrained baseline writes  $\sim 20,000$ -character traces and hit the ceiling on 43.8% of its cued samples, against 21.7% for the reasoning-preserving arm; a truncated trace cannot emit a parseable answer, so the behavioural metrics were computed on a subset that discarded nearly half of one arm and a fifth of the other. We therefore re-ran both arms at 32k, identical in every other respect (Task L, results/reanalysis/L\_budget\_matched\_32k.\*).

The missingness does close: baseline parseability rises  $56.2\% \rightarrow 73.3\%$  and its truncation rate falls  $43.8\% \rightarrow 24.8\%$ , narrowing the across-arm parseability gap from 22.1 to 9.5 points. Two things follow, and neither favours us.

First, the behavioural null is not a truncation artifact. The paired influence interval is essentially unchanged in width ( $0.217 \rightarrow 0.221$ ) even though  $n$  rises from 83 to 95:  $\Delta = +0.032$ , 95% CI  $[-0.074, +0.147]$ ,  $p = 0.71$ . Budget was the wrong lever; only more questions would resolve it.

Second, the acknowledgment effect *weakens* at the fairer budget. Baseline acknowledgment moves  $25.0\% \rightarrow 23.6\%$  while the trained arm moves  $14.4\% \rightarrow 16.6\%$ , so the paired drop falls from  $\Delta = -0.110$  ( $p = 0.005$ ) to  $\Delta = -0.080$   $[-0.153, -0.013]$ ,  $n = 150$ ,  $p = 0.043$ . The gap narrows from 10.6 to 7.0 points. We report the budget-matched figure as the headline because it is

Table 4: Per-text-cue acknowledgment, baseline R1-7B vs. W2SR (paired on question id). Ack drops on every text cue, and each cue’s discordant cells are one-directional (W2SR-only ack  $\leq 1$ ). The two non-text cues (`visual_squares`, `xml_metadata`) sit at 0% on both conditions and are omitted. Per-cue  $p$ -values are exploratory and uncorrected for multiplicity.

| cue                | baseline R1-7B | W2SR (1.5B) | paired $\Delta$ ( $n$ , disc. base/W2SR, $p$ ) |
|--------------------|----------------|-------------|------------------------------------------------|
| stanford_professor | 15/32 = 46.9%  | 3/38 = 7.9% | -0.40 (30, 12/0, $5 \times 10^{-4}$ )          |
| grader_hack        | 7/32 = 21.9%   | 1/38 = 2.6% | -0.20 (30, 7/1, 0.07)                          |
| insider_info       | 18/32 = 56.2%  | 2/38 = 5.3% | -0.50 (30, 15/0, $6 \times 10^{-5}$ )          |

the specification in which neither arm is advantaged, and we note plainly that it is significant only marginally.

The same re-run also corrects a capability number, and the correction runs against us. Baseline’s uncued GPQA accuracy was substantially a budget artifact: 0.275 at 8k against 0.425 at 32k on the all-items convention. But the reasoning-preserving arm, scored on the *same* 40 uncued questions at the same budget, reaches only 0.250. Paired, that is  $\Delta = -0.175$ , 95% CI  $[-0.350, +0.000]$ ,  $p = 0.092$ , discordant 10/3: marginal, but an interval whose upper bound touches zero and does not exclude real harm. The 8k ordering (0.375 arm against 0.275 baseline, Table 2) is reversed. This is not a truncation effect — at 32k the arm is *more* parseable than baseline on these items (31/40 against 27/40) and still less accurate, so it answers more often and gets more of them wrong. We therefore withdraw any reading in which distillation preserves or improves accuracy on this substrate: at a budget that does not advantage either arm, the intervention costs accuracy on the very eval the monitorability numbers are measured on, and the acknowledgment drop should be read against that. Truncation is not eliminated even at 32k — 24.8% of baseline’s cued samples still exhaust it — so we do not claim this comparison is fully budget-saturated, only that it is fairer than the 8k one.

## 6 Discussion

Math-CoT SFT does not preserve monitorability, and where it is measurable, it hurts. The two substrates we look at are consistent in opposite directions relative to what an observer would conclude from Meek’s composite measure. On the floored instruct student, Meek’s composite “monitorability” (the mean of verbosity and faithfulness scores defined in [2]) rises after distillation almost entirely via verbosity. The model looks more monitorable to that measure without being more monitorable to the cue test. On the reasoning student, verbal acknowledgment falls without any matching reduction in the cue’s behavioral pull — strengthened on the answer-only arm, not resolvable either way on the reasoning-preserving one. Compression alone predicts most of that raw collapse; what a compression-only account does not predict is the cue-stratified matched-length gap (§5.1; the influenced = 1 variant of that cut is the paper’s most fragile number and nothing here rests on it) — nor the Task K rerun, which removes most of the compression by fixing the supervision and still finds acknowledgment significantly below baseline. The safety-relevant component gets worse, not better.

The warning is sharp. An accuracy-only certification of any CoT-SFT student trained on MATH-style traces, including same-strength self-distillation rather than just weak-to-strong as in W2SR, *could* pass a model whose chain-of-thought has become markedly less revealing. We are careful with the modal verb: no arm in this paper actually clears its capability gate, so we demonstrate the monitorability loss and the inadequacy of the metric, not a certification that passed in practice. and a monitorability check that rewards elaborate-looking CoT misses the same failure. The narrowness of the W2SR accuracy claim itself (in-family teacher and a base student, Section 4) sharpens this further. The certified guarantee is narrow on both axes.

## 7 Limitations

**The headline is marginal at a matched budget.** At the 8k budget used throughout most of this paper the acknowledgment drop is comfortable ( $p = 0.005$ ); at the 32k budget that stops advantaging the trained arm it is  $p = 0.043$  (§5.3). We report the latter. A reader should treat the effect as directionally consistent across both specifications and statistically marginal in the fairer one, and should not read the 8k figure as the estimate. The behavioural half is a wide null under both budgets,

609 and the budget re-run establishes that widening the generation cap does not narrow it — the binding  
610 constraint is the number of questions, not the trace length.

611 **The headline rests on a single arm and a single seed.** The reasoning-preserving result (§5.2) is  
612 one training run of one teacher–student pair, scored by the headline judge alone, with no multi-judge  
613 battery on that arm. Its significance is also specification-sensitive:  $p = 0.043$  at the matched budget  
614 we report,  $p = 0.005$  at 8k, and  $p = 0.077$  if we instead restrict the 8k pairs to those where neither  
615 side hit the cap. The direction and magnitude are stable across those cuts, but the evidence is thinner  
616 than the surrounding robustness battery, which was measured on the answer-only arms. Re-running  
617 the strong-teacher and self-distillation arms with the corrected renderer is the obvious next experiment  
618 and we have not done it.

619 **Most of our robustness evidence is attached to the answer-only arms.** As described in Sec-  
620 tion 3.1, the student’s chat template removed the reasoning span from every SFT example in the  
621 original runs, so those arms were trained on the final-answer segments of math CoT traces rather  
622 than on the reasoning. That bounds what they show. Their CoT compression is explained by the  
623 supervision format and should not be read as an emergent property of CoT distillation. Their teacher-  
624 strength invariance is weaker evidence than it looks: with the teacher’s reasoning excluded from the  
625 loss, weak, strong, and self teachers differ only in their final answers, so their near-identical effect  
626 is close to what one would predict, and the invariance claim should be treated as untested under  
627 reasoning-preserving supervision. The multi-judge battery, the MMLU replication, the rubric swap  
628 and the CoT-conclusion analysis all sit on these arms too, so they corroborate the answer-only collapse  
629 directly and the headline only by analogy. What the answer-only arms do establish independently  
630 is that these models retain the cue’s behavioral pull — indeed strengthen it — while their traces  
631 stop reporting it. The certification argument survives on either arm, since answer-only SFT on math  
632 traces is itself a common and cheap post-training recipe, and it is exactly the kind of change an  
633 accuracy-only gate would wave through.

634 **The behavioral comparison suffers severe differential missingness.** This is the sharpest limitation  
635 on the paper and it bounds every influence number we report. Computing whether a cue moved the  
636 answer requires recovering an answer, and the arms differ enormously in how often they emit one: on  
637 the 150 matched (question, cue) pairs behind the answer-only comparison, 44.0% of baseline records  
638 have no recoverable answer against 6.7% of the trained arm’s. The cause is not subtle — the untrained  
639 baseline writes ~20,000 character traces into an 8k-token budget and frequently never concludes.  
640 Two consequences follow. First, the influence estimate is computed on a subset that discards nearly  
641 half of one arm and almost none of the other, and the discarded baseline records disproportionately  
642 contained text matching the cue target, so the direction of the extraction correction (§5) is partly  
643 what discarding them mechanically produces. Second, and more fundamentally, a comparison of  
644 “does the cue move the answer” between a model that answers 92% of the time and one that answers  
645 56% of the time is not clean under any extraction rule: the loose rule buys coverage with records  
646 whose accuracy is at chance, and the strict rule buys accuracy with attrition. We report the strict rule  
647 and disclose the sensitivity. The fix is experimental rather than analytical — the evaluation needs  
648 a budget large enough that both arms finish — and §5.3 now reports that experiment: at 32k the  
649 parseability gap narrows from 22.1 to 9.5 points, the behavioral interval does not narrow at all, and  
650 the acknowledgment effect weakens. Treat the acknowledgment result as the paper’s evidence and  
651 the behavioral result as directional.

652 **Answer extraction is a measured quantity, not a given.** Every behavioral number depends on  
653 recovering a final answer letter from a completion, and the arms differ enormously in how often  
654 they emit one within the generation budget (§5). We tightened the extractor to require an explicit  
655 answer marker after finding that a looser pattern was recovering cue restatements from unfinished  
656 baseline traces; that correction raised every reported effect. It also leaves the untrained baseline with  
657 no parseable answer on half its uncued items, which is why we decline to rest capability claims on  
658 uncued GPQA. A reader should treat the behavioral metrics as conditional on an extraction rule that  
659 we have documented and can be varied (LOOSE\_FALLBACK\_IS\_AN\_ANSWER in the released pipeline  
660 reproduces the pre-correction numbers).

661 The instruct arm only motivates the substrate choice. The faithfulness claim rests on the reasoning  
662 student. The matched-length residual depends on the cut we take, and we report all three. Pooled

over all five cues the mid-length gap is closed (Task H bin 3, 17.5% vs 15.1%, Fisher  $p = 0.80$ ) and the long-length gap is marginal (bin 4: OR  $\approx 2.2$ , two-sided  $p = 0.14$ ); restricted to the three text cues, which removes a large across-arm imbalance in cue composition within bins, the gap is present at both lengths ( $p = 0.068$  and  $p = 0.016$ ) but on a W2SR side that does not share baseline’s question pool, and the question-matched versions keep the effect size while losing significance; on the influenced = 1 subset the mid-length gap is 70.0% vs 21.1%, OR  $\approx 8.75$ ,  $p = 0.003$ , on  $n = 10$  baseline records and conditioning on a post-treatment variable. Only the cue-stratified cut avoids both the composition confound and the post-treatment conditioning, and it is the strongest of the three, but it is supporting evidence only; none of them carry the argument on their own, which is why §5.2 exists. Compression accounts for a large share of the raw drop in absolute ack rates. We do not claim it is the sole mechanism, and we do not claim it is absent.

The degradation is shown within the Qwen R1-distill lineage, under weak, strong, and same-strength (self) teachers. A same-recipe cross-family test on R1-Distill-Llama-8B (self-distillation, matched LoRA config, 8k generation budget, same MATH L3 to L5 problem pool) fails the capability gate cleanly: GPQA accuracy goes from 50.0% to 22.5% (−27.5 pp at 0% empty completions on a verified-warm endpoint), at about  $10.8\times$  CoT compression (15,420 to 1,424 chars median). This is substrate sensitivity, not infrastructure noise: the recipe that holds Qwen-7B capability while compressing about  $14\times$  over-compresses Llama-8B and craters accuracy, putting the cross-family Self-B arm in the over-compression confound regime, so we do not run its monitorability eval. Cross-family generalization of the dissociation therefore remains open: at matched recipe, Llama fails the gate before the dissociation question is measurable.

The MMLU replication covers five STEM subjects; broader coverage, including humanities and social sciences, is untested, and the per-subject cells are small (30–40 cued samples each) even though the pooled comparison is well powered. Other limits: LoRA rather than full SFT; MATH-L3 to L5 SFT data only, with other domains untested; and judge robustness checked against two alternative families (gemini-2.5-pro, kimi-k2-0905) rather than a full panel. Absolute acknowledgment rates should be read as judge-dependent in any case [15]; what replicates across our three judges is the direction and the one-directional discordant pattern, not the level.

The influenced = 1 cut that carries the matched-length residual conditions on a post-treatment variable (whether the cue moved the answer), and it is one cell of a two-stratum by five-bin grid reported without multiplicity correction, with  $n = 10$  on the baseline side. We report it because it is the subset monitorability is about, but it is the most fragile number in the paper and should be treated as a well-motivated exploratory finding rather than a confirmatory test. The Task K rerun (§5.2) now provides independent, intervention-based support for the same conclusion — a suppression component beyond compression — without conditioning on a post-treatment variable, though that rerun carries its own caveats (one arm, one seed, single judge).

Finally, the analysis plan was preregistered before any trained student existed (PREREGISTRATION.md). Several deviations should be noted. The preregistered primary estimand was W2SR — strong-teacher control, which is null here ( $p = 0.09$ ); our headline is the preregistered secondary estimand, W2SR — baseline. Under the preregistered decision rules this is the H0 case, “no weak-supervision-specific effect, but an SFT-on-CoT effect that is itself reportable,” which is what we report. A more consequential deviation: the preregistration’s validity gate required the W2SR student’s held-out MATH Pass@1 to exceed baseline by at least 5 absolute points (or recover at least 30% of the baseline-to-teacher gap) for condition-2 monitorability numbers to count, and the only committed gate measurement for the W2SR-weak arm — taken when it was re-gated alongside Task K — fails both branches (0.550 vs 0.695, with a repetition-loop flag; the weak teacher sits below the student’s baseline, so the gap-recovery branch does not apply), as does the Task K rerun’s own gate report (format-valid 0.80, gain +3.5pp). Under a strict reading of the prereg, the reasoning-arm monitorability numbers are therefore reported despite a failed capability gate. We report them because the paper’s claims do not rest on a capability gain — the finding is the collapse, measured on a substrate (GPQA) where the arm’s accuracy is preserved — but a reader holding us to the preregistered gate should treat condition 2 as failing its validity criterion and weight the reasoning-arm results accordingly. The prereg also called for judge–human agreement on a hand-labeled set before trusting the acknowledgment labels; we substituted cross-judge agreement across three model families and a within-judge rubric swap, which is weaker on the human-validity axis. Smaller deviations: the preregistered combined monitorability score is not reported for the

reasoning-student arms (we report its components separately), the per-cue comparisons in Table 4 are exploratory and uncorrected, and the optional mixed-effects robustness model was not run.

## 8 Conclusion

Fine-tuning an R1-distill student on a weaker model’s math chain-of-thought, with the reasoning genuinely in the loss, cuts the trace’s acknowledgment of a planted cue from 23.6% to 16.6% at a matched 32k generation budget (paired  $\Delta = -0.080$ ,  $n = 150$ ,  $p = 0.043$ ; the same comparison at 8k, where only the baseline is routinely truncated, reads 25.0% to 14.4% at  $p = 0.005$ ) and at held-out MATH accuracy that does not fall, and with no detectable change in the cue’s pull on the answer (paired  $\Delta = +0.048$ , CI  $[-0.060, +0.157]$ ,  $p = 0.52$  — an underpowered null on a measurement degraded by differential answer emission, not a demonstrated invariance). Because the retrained traces are only  $\sim 27\%$  shorter than the untrained model’s, the drop is not a by-product of the extreme compression that answer-only supervision produces; and because the manipulation is the training data rather than a filter on the outcome, the result does not condition on anything measured after treatment.

A second arm bounds how much of the effect trace length alone can produce. Due to a chat-template defect, an earlier run supervised the same student on the final-answer segments of the same traces; it compresses CoT  $14\times$  and drives acknowledgment to 3.2%, so answer-only supervision reproduces a far larger collapse than reasoning-preserving supervision does. Length is nonetheless not the whole story even there: at matched length and matched cue composition the gap survives across constructions, though every one of them is fragile — the question-matched versions rest on 7 and 8 records (§5.1) — and the MMLU replication reproduces the full collapse at  $2.8\times$  compression rather than GPQA’s  $14\times$ . The collapse on that arm is also not specific to weak-to-strong distillation — a same-strength self-distillation control reproduces it at indistinguishable magnitude — and it survives a within-judge rubric swap (tightening to 9.4%  $\rightarrow$  0.0%) and two out-of-family judges. A same-recipe cross-family attempt on R1-Distill-Llama-8B fails the capability gate ( $-27.5\text{pp}$  accuracy at  $10.8\times$  compression), so the cross-family scope of the dissociation remains open. Separately, W2SR’s accuracy gain is real but conditional, requiring an in-family teacher and a base student.

Certifying any CoT-SFT student on accuracy alone, or on how elaborate its CoT looks, is insufficient and potentially misleading. The cue-acknowledgment component of monitorability must be measured directly.

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